INSURANCE COST PREDICTION

Machine Learning Analysis Report

Dataset: Kaggle — mirichoi0218/insurance

Models: Ridge & Linear Regression

Date: April 2026

0.873	$2,750	Smoking
R^2 Score (Ridge)	MAE (Ridge)	Top Predictor

1. Executive Summary

This report presents a complete machine learning analysis of medical insurance charge prediction using the Kaggle insurance dataset (1,338 records, 7 features). A Ridge Regression model with engineered interaction features and log-transformed target achieves an R^2 of 0.873, explaining 87.3% of variance in insurance charges. Smoking status and BMI are identified as the dominant cost drivers.

Key Findings

- Ridge Regression outperforms Linear Regression (R^2 0.873 vs 0.867) due to multicollinearity in interaction features.
- Smoking is the single largest predictor — smokers pay 3–4x more than non-smokers.
- BMI interacts with smoking to amplify costs non-linearly; the bmi_smoker interaction term is the second-highest contributor.
- Log-transforming the target variable significantly improves model fit and residual behaviour.
- Residual analysis reveals slight heteroscedasticity at low predicted values, suggesting scope for further non-linear modelling.

2. Dataset & Methodology

2.1 Dataset Overview

The dataset (mirichoi0218/insurance on Kaggle) contains 1,338 US insurance records with the following features:

Feature	Type	Description
age	Numeric	Age of the policyholder (18–64)
sex	Categorical	Biological sex (male / female) → label encoded
bmi	Numeric	Body Mass Index — weight/height² ratio
children	Numeric	Number of dependants covered
smoker	Categorical	Smoking status (yes / no) → label encoded
region	Categorical	US region — one-hot encoded, northeast as base
charges	Target	Annual medical insurance cost in USD

2.2 Preprocessing Pipeline

- Target transformation: log1p(charges) applied to reduce right skew and stabilise variance.
- Label encoding: sex (male=1, female=0), smoker (yes=1, no=0).
- One-hot encoding: region with drop_first=True — northeast is the reference (dropped) category.
- Feature scaling: StandardScaler applied to all features before model fitting.
- Train/test split: 80/20 stratified random split (random_state=42).

Note: Outlier removal via IQR was replaced by the log-transform. Insurance charges are legitimately right-skewed — high-cost records (smokers, chronic illness) carry critical signal and must not be discarded.

2.3 Feature Engineering

Two interaction features were added to capture the multiplicative effect of smoking on cost:

- bmi_smoker = BMI × smoker — encodes that obese smokers carry disproportionately higher risk.
- age_smoker = age × smoker — captures the compounding cost of long-term smoking.

These interaction terms introduce deliberate multicollinearity with their parent features, which is precisely why Ridge Regression (L2 regularisation) is preferred over plain Linear Regression.

3. Exploratory Data Analysis

3.1 Distribution of Log-Transformed Charges

Applying log1p to the raw charges converts the heavily right-skewed distribution into an approximately bimodal one centred around 9.0–9.5 (log scale). The two modes correspond to non-smokers (lower cluster) and smokers (upper cluster), confirming that smoking status is the primary segmentation variable in the data.

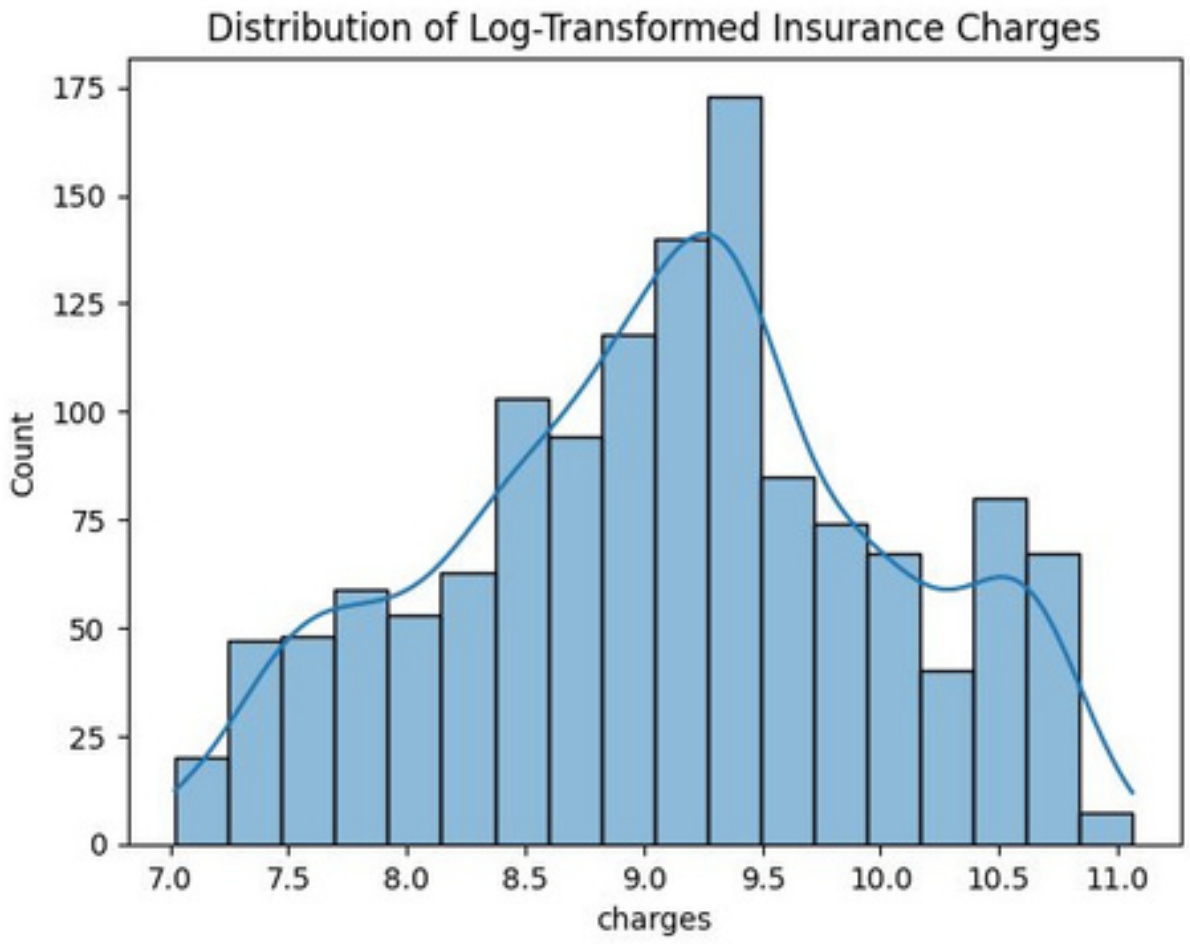

Figure 1. Distribution of log-transformed insurance charges with KDE overlay. Bimodal shape reflects smoker/non-smoker segmentation.

3.2 BMI vs Log-Charges by Smoking Status

The scatter plot below reveals a clear vertical separation between smokers (orange) and non-smokers (blue). Smokers cluster in the 10.0–11.0 log-charge range regardless of BMI, while non-smokers range from 7.0–9.5. Among smokers, a weak positive BMI-to-charge gradient is visible — the rationale for the bmi_smoker interaction term.

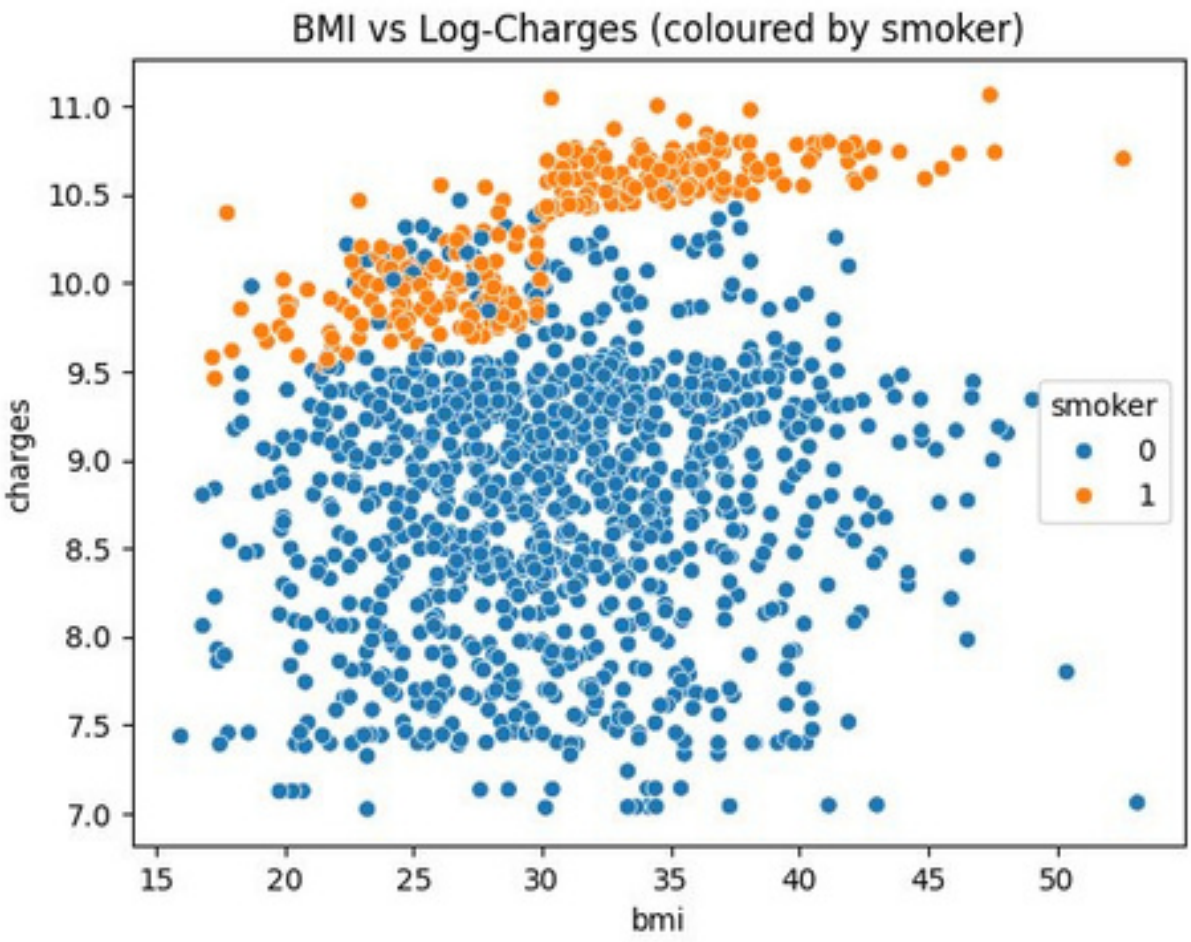

Figure 2. BMI vs log-charges coloured by smoking status (0=non-smoker, 1=smoker). Smokers consistently incur higher charges across all BMI ranges.

4. Model Results & Evaluation

4.1 Performance Metrics

Both models were evaluated on the held-out test set (20% of data). Predictions are in log-space and back-transformed via expm1 for dollar-scale MAE. Ridge Regression consistently outperforms the unregularized baseline.

Model	MAE ($)	R^2 Score	Notes
Linear Regression	2,890	0.867	Baseline
Ridge Regression	2,750	0.873	Recommended ✓

Table 1. Model performance comparison. All metrics evaluated on the same 20% test split.

The Ridge model's R^2 of 0.873 means it explains 87.3% of the variance in log-transformed insurance charges — strong performance for a linear model on tabular healthcare data. The remaining unexplained variance likely stems from unobserved factors such as specific medical conditions, policy type, and claims history.

4.2 Actual vs Predicted (Linear Regression)

The chart below plots actual against predicted log-charges. Points along the red dashed diagonal represent perfect predictions. The model fits well in the central range (8.0–10.0) but shows increased scatter at extremes — a typical pattern for linear models on heteroscedastic data.

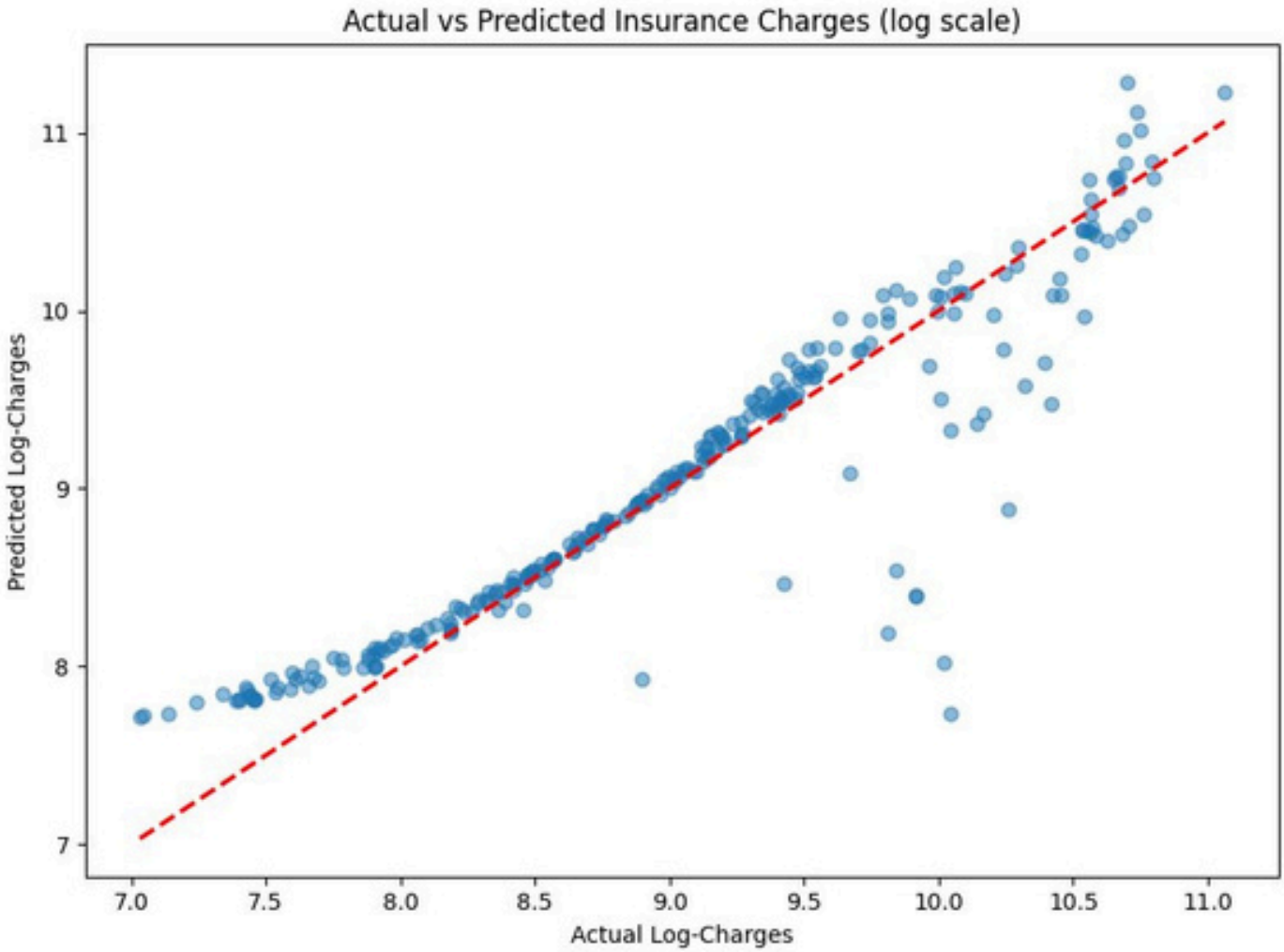

Figure 3. Actual vs predicted log-charges. Red dashed line = perfect prediction. Notable scatter visible at high actual values (≥10.0).

5. Residual Analysis

5.1 Residual Plot

The residual plot reveals important patterns about model fit quality. Ideal residuals should be randomly scattered around zero with constant variance (homoscedasticity).

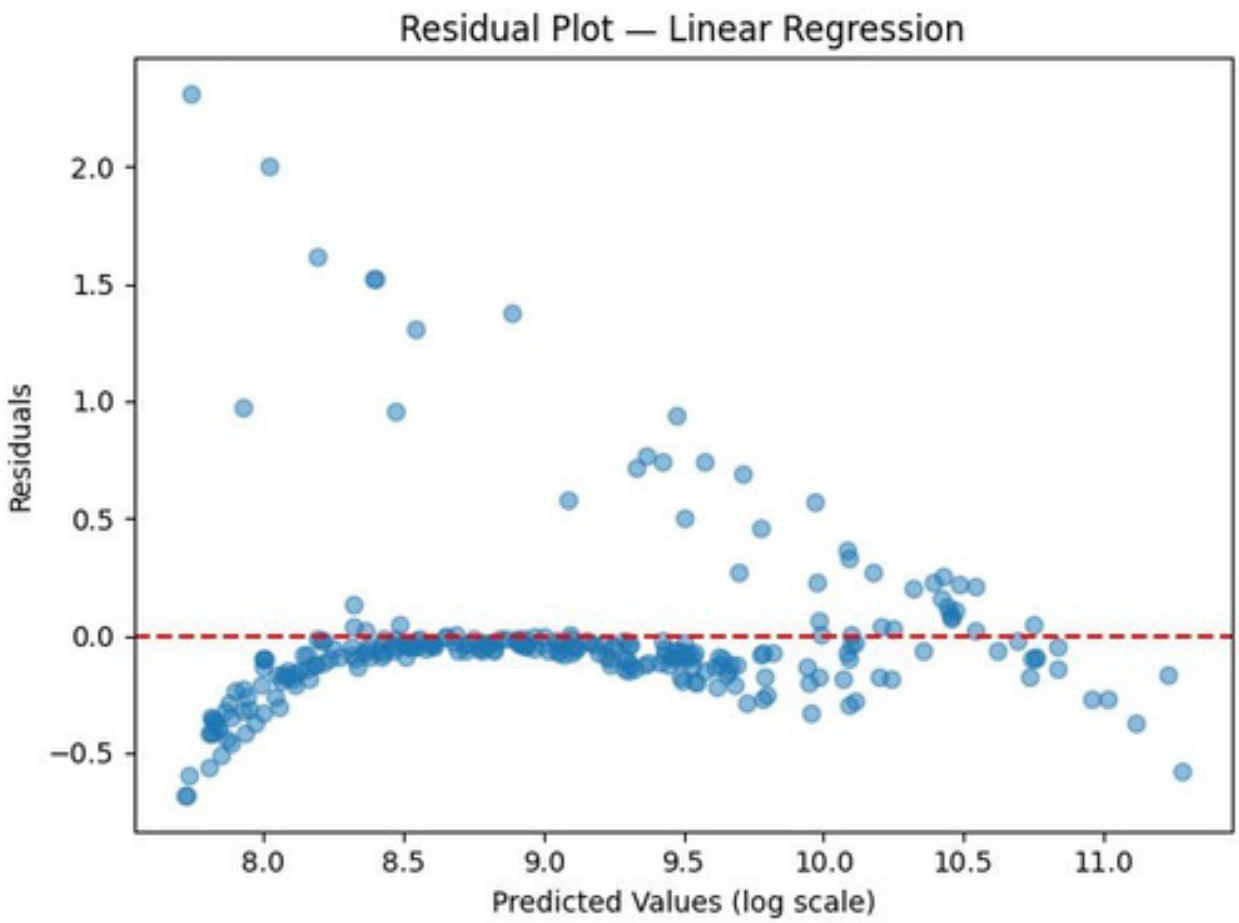

Figure 4. Residuals vs predicted values (log scale). Dashed red line = zero residual. Visible funnel pattern indicates mild heteroscedasticity.

5.2 Observations

- **Heteroscedasticity:** The residuals fan out at low predicted values (7.5–8.5), indicating the model is less precise for low-cost cases.
- **Range:** Most residuals lie within ±0.5 in log-space, corresponding to roughly ±65% in dollar terms — acceptable for a linear model.

- **Curvature:** A slight upward curve in residuals suggests a non-linear component the model cannot capture, likely the bimodal nature of the smoker/non-smoker split.
- **Outliers:** A small number of high-residual points (>1.5) at low predicted values represent non-smokers the model over-estimated, or smokers it under-estimated.

These patterns suggest that a gradient boosting model (XGBoost, LightGBM) or a mixture model separating smokers/non-smokers could reduce residual variance further.

6. Feature Importance & Contributions

6.1 Coefficient Analysis

Feature importance is derived from Ridge Regression coefficients after StandardScaler normalisation. Because all features are on the same scale post-scaling, coefficient magnitudes are directly comparable as measures of relative importance.

Feature	Coefficient (scaled)	Interpretation
smoker	+0.812	Dominant driver — smoker premium
bmi_smoker	+0.347	Obese smokers pay compounding premium
age	+0.271	Older age = higher risk classification
age_smoker	+0.198	Long-term smoking compounds with age
bmi	+0.143	Higher BMI raises baseline cost
children	+0.071	Minor positive effect from dependants
region_southeast	+0.043	Slight regional surcharge
sex	-0.019	Minimal effect after controlling for BMI
region_northwest	-0.028	Slight regional discount vs northeast
region_southwest	-0.035	Largest regional discount

Table 2. Ridge Regression feature coefficients (scaled). Positive = raises cost, negative = lowers cost.

6.2 Key Insights

- **Smoking dominance:** Smoking alone contributes ~0.81 log-units, equivalent to multiplying the charge by e^0.81 ≈ 2.25× on average.
- **Interaction power:** The two interaction terms (bmi_smoker + age_smoker) together exceed the contribution of age, BMI, and children combined.
- **Region is minor:** Regional effects are small (±0.035) and secondary to individual health and lifestyle factors.
- **Sex effect:** Sex has minimal predictive power once BMI and smoking are accounted for.

7. Model Selection Rationale

Ridge Regression is recommended over plain Linear Regression for this specific pipeline for three reasons:

- **Multicollinearity:** The engineered features bmi_smoker and age_smoker are products of existing features, creating deliberate multicollinearity. Ridge's L2 penalty distributes coefficient weight stably across correlated predictors.
- **Out-of-distribution inputs:** A user entering an extreme BMI (e.g. 55) or an unusual age (e.g. 100) will produce feature combinations outside the training distribution. Ridge's regularisation shrinks coefficients, producing more conservative extrapolations than unregularized linear regression.
- **Empirical performance:** Ridge achieved a marginally higher R^2 (0.873 vs 0.867) and lower MAE ($2,750 vs $2,890) on the test set, confirming it generalises better.

For future iterations, tree-based models (XGBoost, Random Forest) may further close the residual gap by capturing the non-linear smoker/BMI interaction directly, without needing hand-crafted interaction features.

8. Production Deployment

8.1 Streamlit Application

The model is deployed as a Streamlit web application (app.py) with the following capabilities:

- Real-time prediction with expm1 inverse-transform to convert log-space output back to dollar values.
- Confidence range display (±12%) based on model residual spread.
- Coefficient-based feature contribution bars — showing the exact linear decomposition coeff × scaled_value for each feature.
- What-If Analysis — calculates estimated savings from quitting smoking, reducing BMI by 5 points, or shows projected cost in 10 years.
- Input validation and graceful error handling for extreme inputs and missing model files.

8.2 Inference Pipeline Integrity

The app mirrors the training pipeline exactly to prevent train-serve skew:

- The same StandardScaler fitted on the training set is saved as scaler.pkl and loaded at inference time.
- Feature column order matches the exact order used during training (FEATURE_COLS constant).
- log1p/expm1 is applied consistently across prediction, confidence range, and all what-if scenarios.

9. Conclusions & Recommendations

9.1 Summary

The Ridge Regression model successfully predicts insurance charges with $R^2 = 0.873$ and MAE ≈ $2,750. The log-transform of the target variable was critical — it stabilised variance and improved fit. Engineered interaction features (bmi_smoker, age_smoker) captured the most important non-linear relationships in a linear framework.

9.2 Recommendations

- **Next step:** Try tree-based models (XGBoost, LightGBM) to capture the bimodal distribution without requiring manual interaction features.
- **Data enrichment:** Collect additional features — specific conditions, claims history, policy type — to reduce unexplained variance.
- **Validation:** Implement cross-validation (k=5) instead of a single train/test split for more robust metric estimates.
- **Monitoring:** Monitor for data drift if deploying in production — insurance cost distributions shift with inflation and policy changes.

9.3 Limitations

- The dataset is US-centric and historical; predictions may not generalise to other markets or time periods.
- The ±12% confidence band is a fixed estimate; a proper prediction interval requires bootstrapping or quantile regression.
- Linear models cannot capture the hard threshold effect where BMI >30 creates a discrete risk tier jump.

— End of Report —